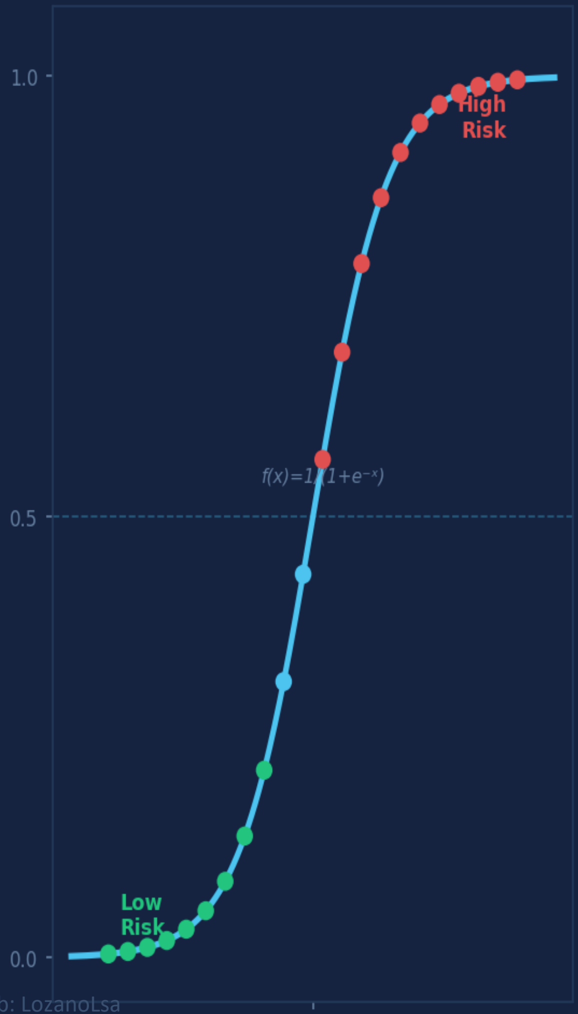
Delays Are Not Random.

What if delays were not random... but predictable?

Logistic Regression · Simulation · Interpretation



1,847
Shipments

9
Features

69.7%
Accuracy

0.662
ROC-AUC

The Real Problem

A delay is not just a late delivery — it's a break in the operational flow.

Costs

Overtime, penalties,
expedited transport

Service

Customer trust erodes
with every late delivery

Operations

Planning cascades fail.
Teams absorb the pressure

There is no single cause. Delays emerge from multiple interacting factors — and that's exactly what ML can model.

In logistics, every delay has a history. The question is whether we're reading it.

From Events to Probabilities

Old Thinking

Delays are isolated events:

"It happened / it didn't."

Binary outcome → binary reaction.
No prediction. No prevention.
Just firefighting after the fact.

→

New Thinking

Delays are probabilistic outcomes:

"How likely is a delay?"

Uncertainty is not noise.
Uncertainty is information.
And information enables decisions.

The right question is not 'Will it be delayed?' — it's 'How likely is a delay, and why?'

The Data Behind the Problem

This dataset doesn't come from a single export — it reflects how data really lives in operations:

ERP	**Orders, priorities, weights, distances** *Commercial & procurement systems*
TMS	**Route type, number of stops** *Transportation & route planning tools*
WMS	**Loading & dock waiting times** *Warehouse & dock scheduling systems*
HR	**Operator experience, departure shifts** *Workforce management & operator master data*

Building a usable table required aligning data across systems. Modeling starts long before training a model.

In a real environment, this dataset doesn't exist in one place — assembling it is half the work.

Why Logistic Regression?

A model your team understands and trusts is more valuable than a black-box with 2% higher accuracy.

Interpretable

Coefficients → odds ratios. Operations teams can read and trust every output.

Probabilistic

Outputs delay probability (0–1), not just binary yes/no. Enables risk scoring.

Fast Baseline

Low compute, easy to deploy. Ideal starting point before adding complexity.

Auditable

No black box. Every factor's contribution is visible and explainable to stakeholders.

Interpretability matters more than complexity — especially when decisions affect people and operations.

Model Results

69.7%	0.662	0.804	91.3%
Accuracy	ROC-AUC	F1 (Delay)	Recall — Delays

Top Delay Drivers (Log-Odds Coefficients)

Route — Urban	+ 0.387
Number of stops	+ 0.375
Loading time (min)	+ 0.204
Dock waiting (min)	+ 0.288
Shipment priority	−0.243
Operator experience	−0.281

Confusion Matrix (370 test shipments)

	Pred: No Delay	Pred: Delay
Actual: No	28 (TN)	90 (FP)
Actual: Delay	22 (FN)	**230 (TP)**

The model captures 91.3% of actual delays — strong recall for an interpretable baseline.

Only 22 real delays out of 206 are missed — each one represents a real operational risk.

From Insight to Action

Urban Routes +0.39	→ Assign experienced operators to urban corridors. Review route planning for high-density zones.
Stop Count +0.38	→ Consolidate stops where possible. Fewer handoffs = less variability = less risk.
Dock Waiting +0.29	→ Schedule loading windows. Stagger arrivals to reduce dock contention at peak hours.
Experience −0.28	→ Match senior operators to complex urban/multi-stop routes — experience is a protective factor.
Priority −0.24	→ Dispatch urgent loads with dedicated fast-lane protocols to reinforce the priority effect.

Not surprises. Just quantified. The model tells you where to act — before the truck leaves the dock.

What-If Simulator — Risk Scorecard

Same model. Different scenarios. Different risks. The model becomes a conversation tool, not just code.

Scenario A — Low Risk

Distance	**120 km**
Stops	**1**
Route type	**Long**
Operator exp.	**10 years**
Priority	**Urgent (1)**
Dock wait	**5 min**

12% delay probability

Scenario B — High Risk

Distance	**50 km**
Stops	**4**
Route type	**Urban**
Operator exp.	**1 year**
Priority	**Standard (0)**
Dock wait	**90 min**

89% delay probability

Probability is a guide, not a verdict. The simulator turns coefficients into operational decisions.

This project is not about predicting the future with certainty.

It is about making uncertainty visible.

Logistic Regression does not provide answers —
it provides better questions.

Feel free to download the full project: dataset, notebook, and simulator.

Change the data. Break the code. Ask different questions.

Because improvement has no finish line.